

ForgeDrive: Bidirectional Cross-Conditioning for Unified Visual-Action Generation in Autonomous Driving

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Abstract

World-model-based autonomous driving endows the model with the ability to understand scene evolution. Yet this promise is undermined by the prevailing imagine-then-act paradigm, which allows errors from the more challenging visual generation stage to cascade into action planning. We introduce ForgeDrive, a unified autoregressive diffusion framework with visual-action cross-conditioning that closes this gap through act-then-imagine paradigm. ForgeDrive factorizes the future as a sequence of per-timestep frame-action pairs, intertwining each action with its corresponding visual observation. During training, we decouple the diffusion timesteps of the two modalities and introduce a UniDiffuser-style noise scheduler to get the ability to infer either modality from its counterpart and deepen understanding of relationships between images and actions. At inference, we propose a novel act-then-imagine inference paradigm, and find that at each step, action generation is a capability internalized during training, requiring no clean future frame as a prerequisite at inference time; instead, the generated action can improve the accuracy of future frame generation, which in turn enhances the quality of the next action. Additionally, we augment each step with future ego-status prediction, further sharpening planning ability. Extensive experiments on NAVSIM demonstrate that ForgeDrive not only unifies driving simulation, planning, and visual odometry into a single model, but also outperforms existing strong planners without any post-training strategy.

Introduction

A central aspiration of autonomous driving is to endow agents with the ability to anticipate the future before acting. World models fulfill this need by learning to predict future states directly from raw video, internalizing environment dynamics to foresee the consequences of potential actions without costly annotation. This combination of predictive capability and data efficiency has made world models increasingly central to autonomous driving.

The key to world-model-based approaches (Zhang et al. 2026; Yang et al. 2026a; Li et al. 2026; Zheng et al. 2025) lies in enabling the model to faithfully capture how the scene evolves, thereby yielding more informed and robust planning decisions. Existing approaches that jointly generate images

and actions can be broadly classified into two paradigms. GPT-style methods (Chen, Wang, and Zhang 2025; Jia et al. 2023; Zhao et al. 2026; Zeng et al. 2026; Li et al. 2025b), as shown in Figure 1 (a), tokenize both future observations and actions, and model them jointly as a unified token sequence via next-token prediction. However, the interaction between the two modalities is mediated solely through causal attention over a flat token sequence, providing only an indirect form of coupling that struggles to capture the fine-grained interplay between ego behavior and scene evolution. Diffusion-based methods (Xia et al. 2026), as shown in Figure 1 (b), first synthesize future frames via a video generation DiT and then condition a downstream trajectory predictor on the resulting features, tightening the connection between visual predictions and planned actions. Yet without action guidance, future frame synthesis relies solely on pixel-level motion extrapolation, making it vulnerable to compounding errors that subsequently corrupt trajectory prediction. Furthermore, all of the pipelines lack a feedback mechanism for the planner to rectify such visual inaccuracies. A natural question thus arises: *how can we forge a tighter, bidirectional coupling between future visual states and actions, so that each modality continuously anchors and corrects the other?*

To address this question, we argue that such bidirectional coupling presupposes the ability to infer either modality from its counterpart: recovering actions from visual observations and synthesizing future frames from actions. To this end, we propose **ForgeDrive**, an autoregressive diffusion framework that unifies visual and action generation through cross-conditioning. Rather than committing to all frames in one shot, ForgeDrive builds the future progressively as a sequence of frame-action pairs, thereby endowing the model with the ability to use actions to correct future visual predictions. To equip the model with the prerequisite bidirectional inference capability, we decouple the diffusion timesteps of the two modalities and introduce a UniDiffuser-style noise scheduler (Bao et al. 2023) that enables mixed training across three complementary objectives: inverse dynamics modeling, action-conditioned video generation, and joint video-action prediction. During inference, we observe that action prediction is comparatively easy and largely mastered during training, requiring only a coarse sense of the upcoming state at inference, whereas future frame synthesis is substantially harder and benefits significantly from action guidance. This

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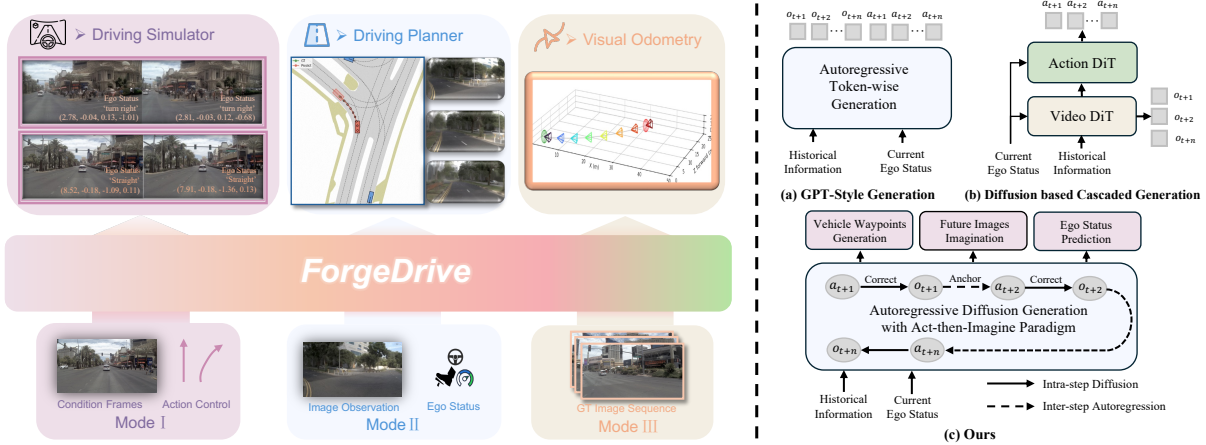


Figure 1: ForgeDrive: a unified autoregressive diffusion framework for autonomous driving. **Left:** ForgeDrive unifies three capabilities—driving simulator, driving planner, and visual odometry—within a single model. **Right:** Comparison of different world-model paradigms. Our model creates a virtuous cycle that progressively tightens both modalities over the horizon: within step, the predicted action corrects the corresponding image generation; across steps, the generated image anchors the next action.

asymmetry motivates an *act-then-imagine* strategy: at each timestep the action is resolved first to correct the subsequent imagination, and the resulting faithful visual context in turn forces the next action to be better informed, creating a virtuous cycle that progressively tightens both modalities over the horizon. Beyond frame-action generation, we further augment each timestep with a prediction of the next-step ego status. This auxiliary objective compels the model to maintain physical consistency between the predicted action and the resulting vehicle dynamics, providing a complementary supervision signal that sharpens trajectory quality.

Owing to the above architectural and strategic designs, ForgeDrive also unifies multiple capabilities within a single model. (1) **Full-state driving simulator:** it can generate future frames and the corresponding ego status conditioned on given observations and a specified action sequence. (2) **Driving planner:** it reasons about plausible future trajectory directly from the current observations and ego status. (3) **Visual odometry:** it recovers the relative pose between frames from an input image sequence. This unified capability set eliminates the need for separate, task-specific models and allows each function to benefit from the representations learned by the others.

In summary, our contributions are as follows:

- We propose **ForgeDrive**, a unified autoregressive diffusion framework that predicts future frames, actions, and ego status for autonomous driving, where multi-objective mixed training deepens cross-modal understanding and unifies driving simulation, planning, and visual odometry within a single model.
- We investigate the causal relationship between future state and action generation, revealing that the counter-intuitive *act-then-imagine* order yields more robust predictions than the prevailing imagine-then-act paradigm.
- Extensive experiments on NAVSIM benchmark demonstrate ForgeDrive achieves strong planning performance

under purely supervised learning, without reinforcement learning or trajectory scoring, outperforming several competitive existing planners.

Related Works

End to End Autonomous Driving

End-to-end autonomous driving (Jiang et al. 2024; Jia et al. 2025; Yin et al. 2026) aims to directly map raw sensor observations to planning decisions within a unified framework. These methods typically require explicit perception modules to extract intermediate representations. Query-based approaches such as UniAD (Hu et al. 2023b), VAD (Jiang et al. 2023), and SparseDrive (Sun et al. 2025) chain perception, prediction, and planning through learnable queries, producing deterministic trajectory outputs via regression. More recent works adopt generative formulations to capture the multimodal nature of driving: GenAD (Zheng et al. 2024) treats trajectory prediction as a latent generative process, DiffusionDrive (Liao et al. 2025) employs a truncated diffusion model for diverse trajectory sampling, and GoalFlow (Xing et al. 2025) leverages flow matching to generate goal-conditioned trajectories. However, all these methods operate solely on the action modality without modeling future visual observations, forgoing the opportunity to leverage predicted scene context for more informed planning.

World Models for Autonomous Driving

In autonomous driving, world models endow the model with the ability to predict the future, thereby improving the quality of trajectories, similar to the Vision-Language Navigation task (Liu et al. 2026b; Chen et al. 2026). Representative works such as GAIA-1 (Hu et al. 2023a), DriveDreamer (Wang et al. 2024a), Drive-WM (Wang et al. 2024b), and Vista (Gao et al. 2024) focus on generating high-quality future frames, leveraging autoregressive transformers

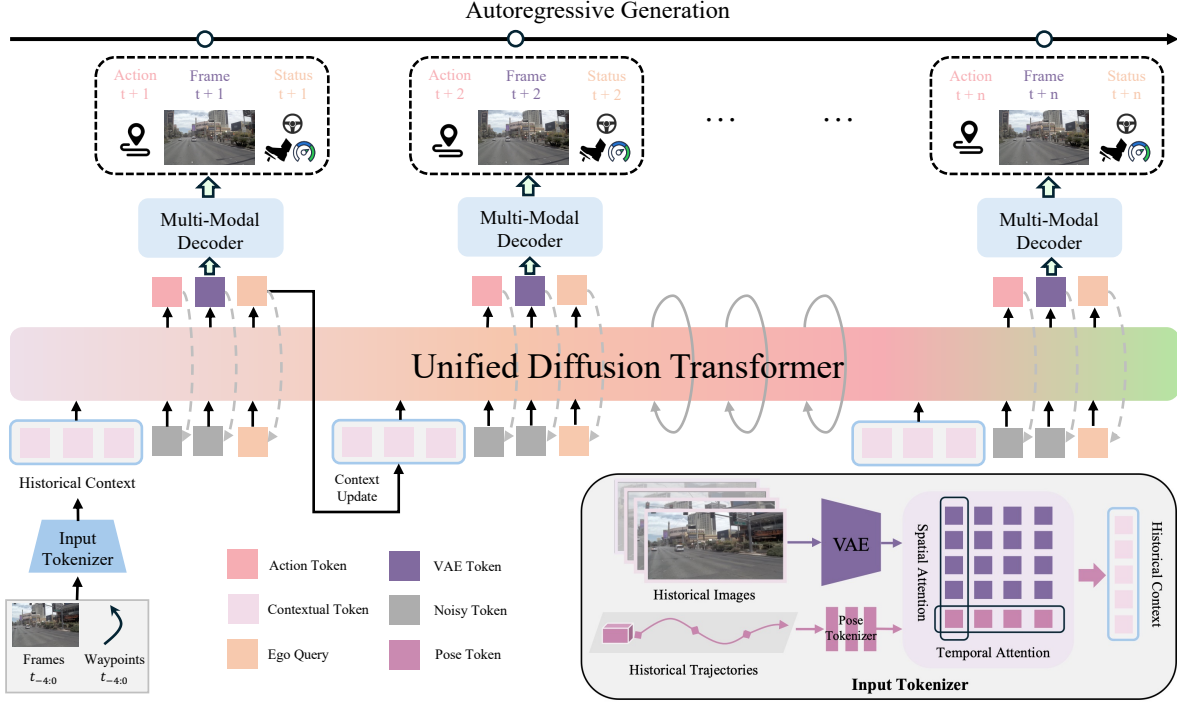


Figure 2: Overall architecture of ForgeDrive. At each autoregressive step, a sliding window of T historical frames with poses and the current ego state is encoded into a context feature via a frozen autoencoder and spatial-temporal attention. The **UniDiT** then denoises image and action tokens under decoupled diffusion timesteps, with full cross-modal attention throughout, and outputs the next frame, action, and ego state in one forward pass. The prediction is fed back into the window to update context for the next step, rolling out the full horizon autoregressively.

or video diffusion models. Recent works bridge video and action generation within a single model: DrivingGPT (Chen, Wang, and Zhang 2025), PWM (Zhao et al. 2026) and ADriver-I (Jia et al. 2023) jointly predict frames and actions via next-token prediction; Epona (Zhang et al. 2025) uses twin parallel DiTs from a shared causal latent; DriveLAW (Xia et al. 2026) unifies both in a shared latent space by serially connecting video and action DiTs. However, these methods either treat the two modalities as a flat token sequence or generate them sequentially without exploiting the interplay between two modalities. ForgeDrive instead decouples the diffusion timesteps of image and action tokens within a single forward pass, creating bidirectional learning signals through cross-conditioning, and reverses the conventional generation order via an act-then-image inference strategy.

Methods

Overview

Given a sequence of T historical camera observations $\{\mathbf{I}_{-T+1}, \dots, \mathbf{I}_0\}$ along with the corresponding ego-vehicle poses and the current ego status $\mathbf{s}_0$, the goal is to predict a sequence of H future frame-action-status tuples $\{(\hat{\mathbf{I}}_1, \hat{\mathbf{a}}_1, \hat{\mathbf{s}}_1), \dots, (\hat{\mathbf{I}}_H, \hat{\mathbf{a}}_H, \hat{\mathbf{s}}_H)\}$. As illustrated in Figure 2, ForgeDrive operates autoregressively along the temporal axis: at each future timestep, the model takes a sliding window of the most recent T frames along with their associated

poses and the current ego status, predicts the next frame-action-status tuple via a Unified Diffusion Transformer, and appends the prediction to the window for the subsequent step.

Model Architecture

Input tokenization. Each raw image $\mathbf{I}_t$ is encoded into spatial tokens $\mathbf{z}_t \in \mathbb{R}^{L \times C}$ by a frozen pre-trained DCAE encoder (Chen et al. 2025), where $L = H' \times W'$. The inter-frame ego-vehicle pose $(\Delta x_t, \Delta y_t, \Delta \psi_t)$ is mapped to pose tokens $\mathbf{e}_t^{\text{pose}} \in \mathbb{R}^{L_{\text{pose}} \times D}$ via Fourier feature embedding and learned projectors. The two groups are concatenated per frame to form $\mathbf{h}_t = [\mathbf{e}_t^{\text{pose}}; \mathbf{z}_t] \in \mathbb{R}^{L_{\text{total}} \times D}$. The T -frame sequence, augmented with learnable embeddings, is processed by spatial-temporal blocks. Each block alternates between causal temporal attention and full spatial attention:

$$\tilde{\mathbf{h}}_t^{(l)} = \text{TemporalAttn}(\mathbf{h}_t^{(l)}, \{\mathbf{h}_\tau^{(l)}\}_{\tau \leq t}), \quad (1)$$

$$\mathbf{h}_t^{(l+1)} = \text{SpatialAttn}(\tilde{\mathbf{h}}_t^{(l)}), \quad (2)$$

The output of the last condition frame serves as the historical context feature $\mathbf{c} = \mathbf{h}_0 \in \mathbb{R}^{L_{\text{total}} \times D}$. Additionally, the current ego state $\mathbf{s}_t$ is encoded into a single token $\mathbf{e}^{\text{ego}} \in \mathbb{R}^{1 \times D}$ by summing the outputs of an MLP applied to the dynamics and a learned embedding of the navigation command.

Unified Diffusion Transformer. Existing two-stage pipelines employ separate networks for video generation and

trajectory prediction, preventing information flow between the two tasks during denoising. ForgeDrive instead unifies both modalities into a single Diffusion Transformer (DiT).

The UniDiT operates on the concatenation of four token groups, each augmented with a learned token-type embedding:

$$\mathbf{X} = \mathbf{c} \oplus \mathbf{z}_{t_{\text{img}}}^{\text{img}} \oplus \mathbf{z}_{t_{\text{act}}}^{\text{act}} \oplus \mathbf{e}^{\text{ego}} \in \mathbb{R}^{(L_c + L_{\text{img}} + L_{\text{act}} + 1) \times D}, \quad (3)$$

where $\oplus$ denotes concatenation along the token dimension, $\mathbf{z}_{t_{\text{img}}}^{\text{img}}$, $\mathbf{z}_{t_{\text{act}}}^{\text{act}}$ are the noisy image and action tokens at their respective diffusion timesteps.

Following the MMDiT architecture (Esser et al. 2024; Labs et al. 2025), the sequence $\mathbf{X}$ is processed through N_d double-stream blocks and N_s single-stream blocks. All blocks perform joint attention over the full token sequence, enabling unrestricted information exchange across modalities:

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \text{Proj}_{q,k,v}^m(\mathbf{X}^{(l)}), \quad (4)$$

$$\mathbf{A} = \text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \quad (5)$$

where $m \in \{\text{img}, \text{act}, \text{cond}\}$ indexes the modality, and Proj^m denotes per-modality projections in double-stream blocks (shared in single-stream blocks). Each modality stream is independently modulated via AdaLN (Peebles and Xie 2023) conditioned on its corresponding timestep embedding, allowing the network to adapt its processing to the noise level of each modality separately. The UniDiT thus learns to estimate the joint conditional velocity field while simultaneously predicting the next-step ego status from the ego token:

$$(\hat{\mathbf{v}}_{\text{img}}, \hat{\mathbf{v}}_{\text{act}}, \hat{\mathbf{e}}_{t+1}^{\text{ego}}) = \text{UniDiT}_{\theta}(\mathbf{z}_{t_{\text{img}}}^{\text{img}}, \mathbf{z}_{t_{\text{act}}}^{\text{act}}, \mathbf{e}_t^{\text{ego}} | \mathbf{c}) \quad (6)$$

Decoding and Output. The decoding process of UniDiT follows a flow matching formulation, iteratively denoising the image and trajectory latent from pure Gaussian noise to the target distribution. The denoised image latent is then decoded by a DCAE decoder to produce the final frame, while the trajectory is mapped to the action space through an MLP head. After the final denoising step, the ego dynamics and driving command are decoded from the ego token’s hidden representation:

$$\hat{\mathbf{s}}_{\text{dyn}}, \hat{\mathbf{s}}_{\text{cmd}} = D_{\text{dyn}}(\mathbf{e}^{\text{ego}}), D_{\text{cmd}}(\mathbf{e}^{\text{ego}}), \quad (7)$$

where D_{dyn} , D_{cmd} are MLP heads. $\hat{\mathbf{s}}_{\text{dyn}} \in \mathbb{R}^4$ represents ego velocity and acceleration prediction, and $\hat{\mathbf{s}}_{\text{cmd}} \in \mathbb{R}^4$ is logits over navigation commands. For autoregressive generation, rather than feeding back these decoded outputs, we directly recycle the latent representations: the denoised image latent is appended to the context window as updated observation, the ego token hidden state is forwarded as the current ego status, and the predicted action is accumulated into the global trajectory.

Training and Inference Schedule

Bidirectional coupling between visual predictions and actions requires the model to not only denoise both modalities jointly, but also to recover each modality conditioned on the

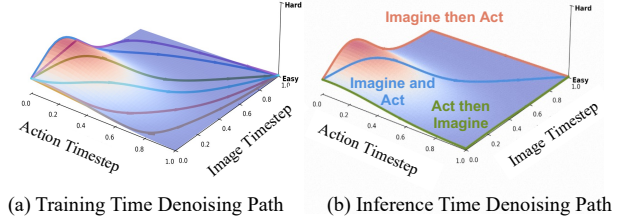


Figure 3: Denoising trajectories during training and inference. Training: we independently vary the diffusion timesteps of the two modalities to cover diverse denoising conditions. Inference: we explore three denoising schedules with different modality resolution orders.

other, thereby enabling one modality to refine and correct the other. This capability is established at training time and operationalized at inference time through a unified design we term cross-conditioning. The training phase addresses the question of *how to equip the model with the ability to infer either modality from its counterpart*; the inference phase addresses the complementary question of *how to best leverage this ability to achieve robust mutual correction during autoregressive rollout*.

Training. To equip the model with the ability to infer either modality from its counterpart, we design a training scheme that exposes the model to diverse cross-modal generation conditions within a single unified framework. Inspired by the multi-modal diffusion framework of UniDiffuser (Bao et al. 2023), as shown in Figure 3 (a), we decouple the diffusion timesteps of the two modalities and sample t_{img} and t_{act} independently from $\mathcal{U}(0, 1)$. Under this scheme, the model learns a unified conditional distribution:

$$p_{\theta}(\mathbf{z}^{\text{img}}, \mathbf{z}^{\text{act}} | \mathbf{c}, t_{\text{img}}, t_{\text{act}}), \quad (8)$$

which subsumes three complementary learning objectives as special cases depending on the sampled timestep pair:

- **Inverse dynamics modeling** ($t_{\text{img}} \rightarrow 1, t_{\text{act}} \rightarrow 0$): the image tokens are nearly clean while the action tokens are heavily noised, reducing to $p_{\theta}(\mathbf{z}^{\text{act}} | \mathbf{z}^{\text{img}}, \mathbf{c})$ and training the model to recover actions from visual observations.
- **Action-conditioned video generation** ($t_{\text{img}} \rightarrow 0, t_{\text{act}} \rightarrow 1$): the action tokens are nearly clean while the image tokens are heavily noised, reducing to $p_{\theta}(\mathbf{z}^{\text{img}} | \mathbf{z}^{\text{act}}, \mathbf{c})$ and training the model to synthesize future frames guided by the intended action.
- **Joint video-action prediction** ($t_{\text{img}} \approx t_{\text{act}}$): both modalities carry comparable noise, training the model to denoise them simultaneously.

By default, the model is trained with teacher forcing, where ground-truth serve as context for subsequent autoregressive steps. However, at inference time the model must condition on its own imperfect predictions, creating a discrepancy that can lead to compounding drift. To bridge this gap, we adopt scheduled sampling: with a probability that warms up linearly during training, the model’s own predictions replace

the ground-truth values in the context window, gradually acclimating the model to autoregressive self-conditioning.

Inference. The training scheme above equips ForgeDrive with the ability to generate either modality conditioned on the other. The inference phase addresses the complementary question of how to best operationalize this ability for robust autoregressive rollout. Because the decoupled timesteps allow the model to follow arbitrary denoising paths, we can choose the generation order that minimizes error accumulation. A key observation (Table 6) guides our choice: single-step action prediction is largely mastered during training, whereas accurate future state estimation is the primary bottleneck that benefits significantly from explicit action guidance. This motivates a cascaded *act-then-imagine* schedule, as shown in Figure 3 (b). At each autoregressive step, the action is denoised first ($t_{\text{act}}: 0 \rightarrow 1, t_{\text{img}}=0$) and then frozen as a condition ($t_{\text{act}}=1$) while the image tokens are denoised ($t_{\text{img}}: 0 \rightarrow 1$). The resolved action corrects the subsequent imagination, whose improved fidelity in turn strengthens the prior for the next autoregressive step, forming the mutual correction cycle that cross-conditioning enables. Unlike action-free pipelines that synthesize an entire future segment without guidance and extract actions post hoc, our cascaded schedule sidesteps both compounding visual drift and the information bottleneck of post-hoc action recovery.

Training Objective

ForgeDrive is trained with rectified flow (Liu, Gong, and Liu 2022). The standard velocity-prediction losses are applied to the image and action tokens, denoted as $\mathcal{L}_{\text{img}}$ and $\mathcal{L}_{\text{act}}$, respectively. Additionally, we supervise the predicted next-step ego status with a combined loss that regresses the dynamics (v_x, v_y, a_x, a_y) via MSE and classifies the navigation command $\mathbf{d}$ via cross-entropy:

$$\mathcal{L}_{\text{ego}} = \|\hat{\mathbf{s}}_{\text{dyn}} - \mathbf{s}_{\text{dyn}}\|^2 + \text{CE}(\hat{\mathbf{s}}_{\text{cmd}}, \mathbf{s}_{\text{cmd}}). \quad (9)$$

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{img}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{ego}} \mathcal{L}_{\text{ego}}. \quad (10)$$

where λ_{act} and λ_{ego} are scalar weights balancing the action and ego-state losses against the image loss.

Experiment

Experimental Setup

Dataset and Metrics. We evaluate our method on the NAVSIM benchmark (Dauner et al. 2024; Cao et al. 2025), a high-fidelity simulation dataset derived from nuPlan (Caesar et al. 2021). To comprehensively validate the planning performance and robustness of our approach, we report results across three official splits: NAVSIM v1 navtest, NAVSIM v2 navtest, and navhard. Following the benchmark protocol, we adopt the PDM Score (PDMS) for v1 and the Extended PDM Score (EPDMS) for v2 and navhard as our primary evaluation metrics. To evaluate video generation quality, we utilize Fréchet Video Distance (FVD) (Unterthiner et al. 2018) to assess the statistical realism of predicted frames. Furthermore, we quantify ego status prediction performance using two distinct criteria: Mean Absolute Error (MAE) for ego dynamics and classification accuracy for driving commands.

Implementation Details. Our Forgedrive comprises a total of 3B parameters and adopts an Mixture-of-Transformer (Liang et al. 2024) architecture, where the video generation parameters are initialized from Epona (Zhang et al. 2025). Its underlying Diffusion Transformer is based on FLUX (Labs et al. 2025) pre-trained on nuPlan at 5Hz. To align the newly integrated action planning with these world model priors, the framework is first adapted on nuPlan at 2Hz for 30,000 steps, establishing a stable joint initialization for both modalities before full training on NAVSIM for 30 epochs. Training is performed on 8× NVIDIA H20 GPUs with a total batch size of 16 and a 512×1024 input resolution, processing 2Hz sequences to predict an 4.0-second future horizon. The network is optimized end-to-end via AdamW (Loshchilov and Hutter 2017), employing a learning rate of 1×10^{-4} and a weight decay of 5×10^{-2} . The multi-task loss weights are configured as $\lambda_{\text{act}} = 1.0$ and $\lambda_{\text{ego}} = 0.01$. At inference, we apply 2 denoising steps to the trajectory and 2 denoising steps to the image (4 steps total) for trajectory evaluation, while using 10 denoising steps for the image when evaluating image quality.

Main Results

Planning Capability. Table 1 compares our method with state-of-the-art approaches on NAVSIM v1. Trained solely with supervised learning on a single camera input, without reinforcement learning or learned trajectory scoring, our method achieves a PDMS of 90.2, surpassing traditional end-to-end methods with LiDAR input. Furthermore, compared to world-model-based methods, our model surpasses DriveLaW using a cascaded DiT generation approach by 1.1 PDMS, and outperforms PWM and Uni-World VLA using GPT-style methods by 2.1 and 0.8 PDMS. Table 2 presents results on the NAVSIM v2, which adopts a more challenging evaluation metric. Our method achieves 90.3 EPDMS with no drop from the PDMS score, alongside an excellent lane keeping score of 98.1, demonstrating the high quality of our trajectories. Compared with DreamAD, which also uses Epona as the video generation backbone but employs reinforcement learning, our method surpasses it by 2.6 EPDMS. Compared with the recent strong method Latent-WAM and DriveFuture, we outperform them by 1.0 and 0.4 EPDMS.

Video Generation Capability. Table 3 evaluates video generation quality and long-horizon capability on NAVSIM test split over a 4-second future horizon. We first compare generation fidelity against VQ-VAE (Van Den Oord, Vinyals et al. 2017) methods: our diffusion-based approach achieves an FVD of 69.2, outperforming Uni-World VLA and Driving-GPT by a large margin, confirming that continuous latent diffusion produces substantially more realistic and temporally coherent futures than discrete tokenization. In terms of long-horizon generation, our model matches Epona in supporting 120-second rollouts, far exceeding Vista at 15 seconds and other methods limited to 4 seconds. Beyond video generation, our method further advances over Epona by jointly predicting ego status and action alongside future scene evolution, elevating the system from a video generator to a full-state driving simulator capable of modeling both environmental dynamics and ego behavior over extended horizons.

Methods	Input	NC↑	DAC↑	TTC↑	Comf↑	EP↑	PDMS↑
<i>Traditional End-to-End Methods</i>							
Uniad (Hu et al. 2023b)	C	97.8	91.9	92.9	100	78.8	83.4
TransFuser (Chitta et al. 2022)	C&L	97.7	92.8	92.8	100	79.2	84.0
Hydra-MDP (Li et al. 2024)	C&L	98.3	96.0	94.6	100	78.7	86.5
DiffusionDrive (Liao et al. 2025)	C&L	98.2	96.2	94.7	100	82.2	88.1
<i>World Model Methods</i>							
Driving GPT (Chen, Wang, and Zhang 2025)	SC	98.9	90.7	94.9	95.6	79.7	82.4
FSdrive (Zeng et al. 2026)	SC	98.2	93.8	93.3	99.9	80.1	85.1
Epona (Zhang et al. 2025)	SC	97.9	95.1	93.8	99.9	80.4	86.2
PWM (Zhao et al. 2026)	SC	98.6	95.9	95.4	100	81.8	88.1
WoTE (Li et al. 2025c)	C&L	98.5	96.8	94.9	99.9	81.9	88.3
DriveVLA-w0 (Li et al. 2025b)	SC	98.7	96.2	95.1	100	82.2	88.4
DriveLaW (Xia et al. 2026)	SC	99.0	97.1	96.7	100	81.3	89.1
Uni-World VLA (Liu et al. 2026a)	SC	98.7	96.7	96.1	100	83.2	89.4
Ours	SC	98.9	97.6	96.3	100	83.8	90.2

Table 1: Comparison with state-of-the-art methods on the NavSim v1 benchmark. **Bold** indicates the best result in each column.

Method	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS*↑	EPDMS↑
TransFuser (Chitta et al. 2022)	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7	—
Hydra-MDP++ (Li et al. 2025a)	97.2	97.5	99.4	99.6	83.1	96.5	94.4	98.2	70.9	81.4	—
DriveSuprim (Yao et al. 2026)	97.5	96.5	99.4	99.6	88.4	96.6	95.5	98.3	77.0	83.1	—
ARTEMIS (Feng et al. 2025)	98.3	95.1	98.6	99.8	81.5	97.4	96.5	98.3	—	83.1	—
DiffusionDriveV2 (Zou et al. 2025)	97.7	96.6	99.2	99.8	88.9	97.2	96.0	97.8	91.0	85.5	87.5
DriveVLA-W0 (Li et al. 2025b)	98.5	99.1	98.0	99.7	86.4	98.1	93.2	97.9	58.9	86.1	—
Recogdrive (Li et al. 2025d)	98.3	95.2	98.3	99.8	87.1	97.5	96.6	99.5	86.5	83.6	—
Epona (Zhang et al. 2025)	97.1	95.7	99.3	99.7	88.6	96.3	97.0	98.0	67.8	—	85.1
DreamAD (Yang et al. 2026b)	98.0	97.2	99.5	99.8	87.8	97.4	97.5	98.3	72.4	—	87.7
Latent-WAM (Wang et al. 2026)	98.1	97.3	99.6	99.8	87.7	97.3	97.6	98.1	87.3	—	89.3
DriveFuture (Hong et al. 2026)	98.8	99.1	99.6	99.9	86.6	98.4	96.4	98.3	74.8	86.4	89.9
Ours	98.9	97.6	99.6	99.8	87.2	98.4	98.1	98.4	86.5	86.6	90.3

Table 2: Comparison with state-of-the-art methods on the NavSim v2 benchmark. **Bold** indicates the best result in each column.

Additional Capability. As shown in Table 5, the results demonstrate the model’s excellent capability in predicting ego status. Furthermore, under the IDM mode, the model can accurately recover driving trajectories directly from images, achieving a remarkable Mean ADE of only 0.11 meters. This high-precision tracking and reconstruction performance highlights the model’s potential to function effectively for visual odometry tasks in autonomous driving scenarios.

Ablation Study

Effect of Training Strategy. The ablation study in Table 4 validates the effectiveness of our training strategy. Introducing decoupled timestep scheduling yields substantial gains, with EPDMS improving from 87.2 to 89.7. This demonstrates that endowing the model with the ability to infer either modality from its counterpart can improve its understanding of the correspondence between image and action. Furthermore, incorporating ego status prediction as an auxiliary objective provides additional improvements, raising EPDMS to 90.0 and increasing EC by 2.4 points. This shows that jointly

training with self-motion supervision enhances the model’s scene understanding and decision-making consistency.

Effect of Inference Strategy. Table 6 reports the planning performance of our model under different inference strategies. Among the three paradigms, *Act-then-Imagine* achieves the highest EPDMS on both navhard and navtest, followed by *Imagine-and-Act*, while *Imagine-then-Act* scores the lowest. The same trend holds for FVD. It indicates that action generation quality doesn’t depend on clear future frame guidance, whereas future frame generation benefits from actions to produce more accurate future states. To further investigate the underlying mechanism, we introduce two additional settings. Under the GT Future setting, where the model only generates actions and future frames are directly taken from ground truth, the model reaches a 92.8 EPDMS, demonstrating that high-quality action generation can be learned during training and doesn’t require clean future frames at inference. Conversely, when we completely mask out the image noise during action generation in the *Act-then-Imagine* mode, the

Metric	Epona	WoVoGen	Vista	DrivingGPT	Uni-World VLA	Ours
FVD ↓	82.8	417.7	89.4	142.6	141.8	69.2
Max Duration / FPS	120s / 2Hz	2.5s / 2Hz	15s / 10Hz	4s / 2Hz	4s / 2Hz	120s / 2Hz
Dataset	nuScenes	nuScenes	nuScenes	NavSim	NavSim	NavSim
View	Front	Multi	Front	Front	Front	Front

Table 3: Quantitative comparison of video generation capability. Methods: Epona (Zhang et al. 2025), WoVoGen (Lu et al. 2024), Vista (Gao et al. 2024), DrivingGPT (Chen, Wang, and Zhang 2025), Uni-World VLA (Liu et al. 2026a).

Decouple Timestep	Ego Prediction	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
✗	✗	98.3	96.4	99.6	99.8	87.1	97.4	97.0	98.2	75.3	87.2
✓	✗	98.7	97.4	99.6	99.8	87.9	98.2	98.1	98.2	83.8	89.7
✓	✓	98.9	97.5	99.5	99.8	87.3	98.4	98.0	98.4	86.2	90.0

Table 4: Ablation study on key training strategies. All the experimental results of the above models are obtained through inference in the Imagine-and-Act mode.

Category	Metric	Value
Ego Status	Dynamics MAE	[0.43, 0.03, 0.31, 0.21]
	Command Acc	94.7%
Trajectory	Mean ADE	0.11

Table 5: Ego status and trajectory recovery results on the NAVSIM test split over a 4-second future horizon. Dynamic MAE is reported in the order of (v_x, v_y, a_x, a_y) .

Infer. Strategies	navhard		navtest	FVD↓
	EPDMS ₁ ↑	EPDMS ₂ ↑	EPDMS↑	
Imagine-then-Act	81.4	39.7	89.9	73.3
Imagine-and-Act	82.8	41.2	90.0	73.4
Act-then-Imagine	83.0	42.3	90.3	69.2
Act-then-Imagine [†]	79.3	41.8	89.2	-
GT Future	-	-	92.8	-
Human	-	-	94.5	-

Table 6: Ablation study on inference strategies. [†] denotes applying image mask during trajectory denoising. GT Future means ground-truth future frames are provided.

performance drops below that of *Imagine-then-Act*, further suggesting that the role of future frames for action is not to provide a clear future state, but rather to convey an evolving trend of the future. Hence, compared to the *Imagine-then-Act* paradigm which generates future frames in an action-free manner, the *Act-then-Imagine* paradigm can not only effectively leverage future information when generating actions, but also use actions to correct the generation of future frames, thereby achieving superior planning performance.

Visualization

Figure 4 illustrates the imagined future frames and planned trajectories under different inference modes. We can observe that in the Imagine-then-Act mode, the vehicle drives out of

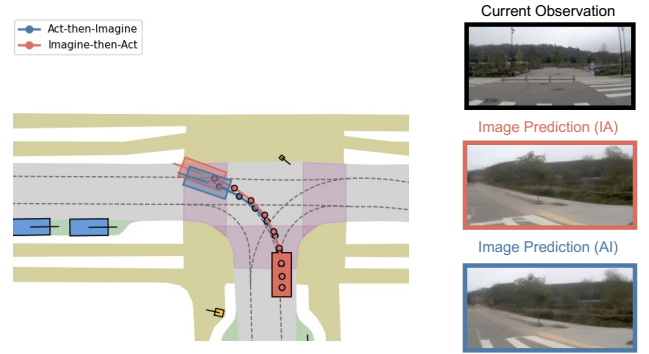


Figure 4: Visualization of different inference modes: IA (Imagine-then-Act) vs. AI (Act-then-Imagine).

the drivable area, whereas in the Act-then-Imagine mode, the imagination can be corrected, keeping the planned trajectory on the road.

Conclusion

We presented ForgeDrive, a unified autoregressive diffusion framework that bridges visual generation and action planning through bidirectional cross-conditioning. By decoupling the diffusion timesteps of image and action tokens during training, our model acquires the ability to infer either modality from its counterpart, unifying driving simulation, planning, and visual odometry within a single architecture. Crucially, we identified and validated an *act-then-imagine* inference paradigm that reverses the conventional generation order, allowing future states to inform action generation while predicted actions simultaneously correct future visual imagination, thereby preserving action quality and avoiding the error cascades inherent in imagine-then-act approaches. Extensive experiments on NAVSIM demonstrate that ForgeDrive achieves strong planning performance without post-training, while simultaneously delivering high-fidelity long-horizon video generation. Future work will explore extending our work to multi-sensor inputs and real-world deployment.

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Supplementary Material

Additional Ablation Studies

Effect of Inference Sampling Steps. Table 7 studies the effect of inference sampling steps on planning performance. With only 2 steps, the model already achieves a strong EPDMS of 89.9, demonstrating that the image-action distribution has been well learned during training. Increasing to 4 steps brings the best overall EPDMS of 90.3, with notable gains in EC (86.5) and TTC (98.4), indicating that slightly more denoising steps improve collision avoidance and temporal consistency. Beyond 4 steps, the trajectory planning performance changes only marginally: at 10 steps, the EPDMS remains competitive at 90.2, while NC and TTC reach their peak; further increasing to 20 steps yields a slight EP improvement but leads to a drop in EC and EPDMS.

Effect of Image Resolution. Table 8 investigates the effect of input image resolution. Increasing the resolution from 256×512 to 512×1024 brings consistent improvements on most metrics, most notably raising the Ego Progress (EP) from 86.6 to 87.3 and the Extended Comfort (EC) from 81.7 to 84.2, which together lift the overall EPDMS from 89.3 to 89.9. The gains in collision-related metrics suggest that higher-resolution inputs provide finer visual details critical for perceiving obstacles and navigating complex scenarios. Meanwhile, the No at Fault Collisions (NC) and Time to Collision (TTC) metrics remain stable or slightly decrease, indicating that the base resolution is already sufficient for basic collision avoidance. These results confirm that a moderate increase in resolution is beneficial for planning robustness, while also justifying our choice of 512×1024 as the default setting.

More Details about Results on Navhard Split

Comparison with other methods. Table 9 compares our method with state-of-the-art approaches on the NAVSIM v2 (navhard split) benchmark. Our method achieves the highest overall EPDMS score of 34.8, surpassing the second-best DriveLaW by a notable margin of 4.2 points. In Stage 1, we attain the best results in NC (97.8), DAC (94.9), LK (96.9), and HC (97.8), with the DAC score exceeding the next best method by 5.8 points, demonstrating a clear advantage in drivable area compliance. In Stage 2, our method continues to lead in TLC (98.7), LK (49.9), and HC (96.6), while remaining competitive on the remaining metrics. In contrast, DiffusionDrive exhibits a sharp drop in safety-related metrics in Stage 2, and MindDrive, despite strong Stage 2 NC and DAC, trails in overall EPDMS due to weaker Stage 1 performance. The consistent strength across both stages enables our method to set a new state-of-the-art on this challenging benchmark.

Comparison between different inference strategy. As shown in Table 10, Act-then-Imagine achieves the best overall EPDMS of 34.8, outperforming Imagine-and-Act (33.9) and Imagine-then-Act (32.4). The advantage is particularly pronounced in safety-critical metrics: on Stage 2, Act-then-

Imagine leads in DAC (70.9), TTC (77.5), and NC (81.5), indicating more collision-free and smoother driving behaviors. This validates the effectiveness of our paradigm—by generating actions first, the model leverages future-state-informed representations to produce reliable trajectories, which in turn serve as corrective signals for subsequent visual imagination, yielding safer and more coherent long-horizon predictions.

Additional Details about UniDiT

Figure 10 illustrates the detailed architecture of our UniDiT model. To better preserve the video generation capability, the main video generation backbone follows the architecture of the existing state-of-the-art work (Zhang et al. 2025; Labs et al. 2025). On top of this, we integrate the input and prediction of actions and future ego status. The overall architecture can be divided into a double-stream block and a single-stream block. In the double-stream block, each input modality is projected through an independent MLP to learn modality-specific features. In the single-stream block, a shared MLP is employed to achieve better cross-modal fusion. Notably, independent timestep modulation is applied between the image stream and the action stream, endowing the model with the ability to infer one modality from the other.

More Qualitative Visualization

Visualization of Planning. Figure 5 illustrates the performance of our method in several scenarios. Our method not only produces accurate future scene predictions but also generates high-quality trajectory planning results. Figure 6 compares our method with the existing approach DriveVLA-w0 (Li et al. 2025b), where it can be observed that our method exhibits better robustness in challenging situations such as turning scenarios.

Visualization of Action-Controlled Video Generation. Figure 7 demonstrates our model’s video generation capability under action control. The ground-truth data is a video of straight driving; our model, given the same conditional frames and a right-turn trajectory as input, can generate the corresponding video for a right-turn scenario.

Visualization of Trajectory Recovery. Figure 8 demonstrates the capability of our model as a visual odometry. Given a video of a turning scenario with large-scale motion, our model can accurately recover the relative pose transformation between consecutive frames.

Visualization of Long Video Generation. Figure 9 demonstrates our model’s ability to generate long videos. Unlike some existing autonomous driving world action models that can only produce a fixed-length video, our model supports the generation of videos with varying lengths. Moreover, when generating longer videos, it does not suffer from blurring and preserves high-quality details of vehicles, roads, and buildings.

Sampling steps	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
2	98.8	97.5	99.7	99.8	87.3	98.3	98.0	98.3	85.7	89.9
4	98.9	97.6	99.6	99.8	87.2	98.4	98.1	98.4	86.5	90.3
10	99.0	97.5	99.6	99.8	87.3	98.5	98.1	98.4	86.1	90.2
20	98.9	97.5	99.6	99.8	87.4	98.4	98.0	98.4	85.9	90.1

Table 7: Ablation study on inference sampling steps. All results are obtained under the cascaded Act-then-Imagine inference paradigm, where the total sampling steps are evenly split between trajectory denoising and image denoising (e.g., 4 steps = 2 steps for trajectory + 2 steps for image).

Resolution	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
256*512	99.0	97.2	99.6	99.9	86.6	98.5	98.0	98.3	81.7	89.3
512*1024	98.9	97.6	99.6	99.8	87.3	98.4	98.2	98.3	84.2	89.9

Table 8: Ablation study on input image resolution. All results are obtained under the Act-then-Imagine setting without ego-status co-training.

Method	Stage	NC	DAC	DDC	TLC	EP	TTC	LK	HC	EC	EPDMS
DiffusionDrive	1	96.8	86.0	98.8	99.3	84.0	95.8	96.7	97.6	79.6	27.5
	2	80.1	72.8	84.4	98.4	85.9	76.6	46.4	96.3	72.8	
MindDrive	1	96.1	86.0	98.8	99.3	83.3	95.6	94.4	97.6	74.7	30.5
	2	82.6	79.1	86.4	98.0	85.3	79.4	49.2	96.5	71.0	
DriveLaW	1	97.3	89.1	99.2	99.6	84.3	97.1	96.2	97.8	67.6	30.6
	2	82.5	67.6	83.5	98.1	84.8	78.5	45.8	96.4	57.3	
Ours	1	97.8	94.9	99.1	99.3	83.7	95.8	96.9	97.8	77.8	34.8
	2	81.5	70.9	84.7	98.7	84.2	77.5	49.9	96.6	68.6	

Table 9: Comparison on the NAVSIM v2 (navhard split) benchmark.

Method	Stage	NC	DAC	DDC	TLC	EP	TTC	LK	HC	EC	EPDMS
Imagine-then-Act	1	98.0	92.9	99.1	99.3	83.9	96.2	96.9	97.8	76.0	32.4
	2	79.9	70.2	83.7	98.6	85.8	76.0	49.1	96.2	65.6	
Imagine-and-Act	1	97.6	94.0	99.1	99.8	83.5	95.5	96.9	97.8	77.8	33.9
	2	80.9	70.8	84.3	98.5	84.8	76.6	49.2	96.3	69.2	
Act-then-Imagine	1	97.8	94.9	99.1	99.3	83.7	95.8	96.9	97.8	77.8	34.8
	2	81.5	70.9	84.7	98.7	84.2	77.5	49.9	96.6	68.6	

Table 10: Detailed results on the NAVSIM v2 (navhard split) benchmark with different inference mode.

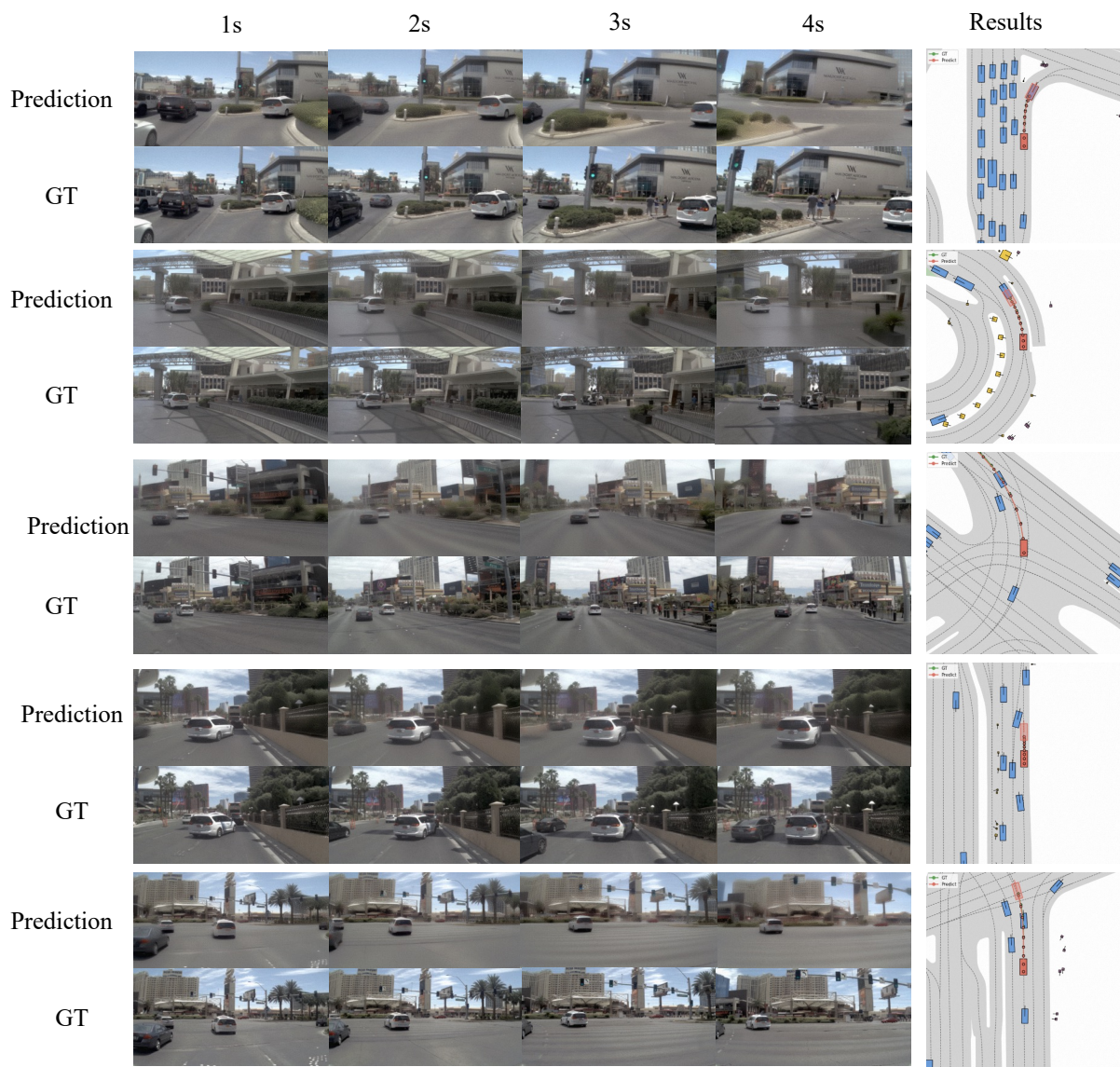


Figure 5: Visualization of planing and future prediction. The results are obtained using two denoising steps for both the image and the action.

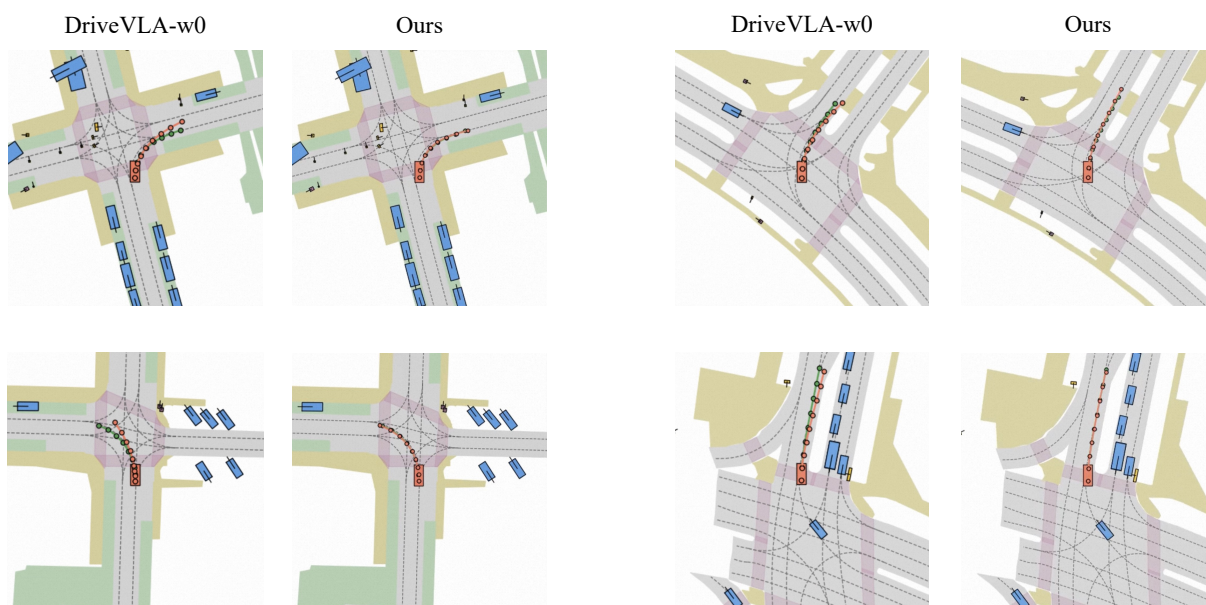


Figure 6: Comparison between our method and DriveVLA-w0(Li et al. 2025b)



Figure 7: Visualization of action-Controlled video generation. The top half shows the ground-truth data, while the bottom half shows the generated video conditioned on a right-turn trajectory.

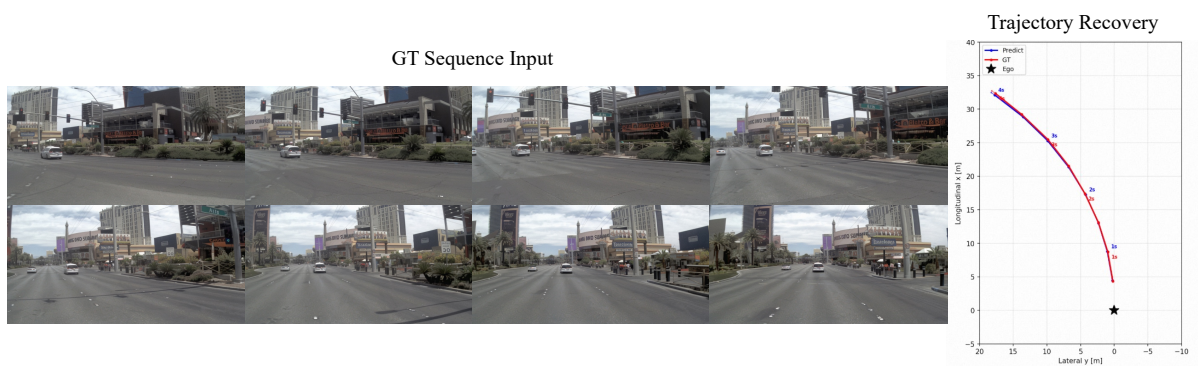


Figure 8: Visualization of trajectory recovery.



Figure 9: Visualization of long video generation.

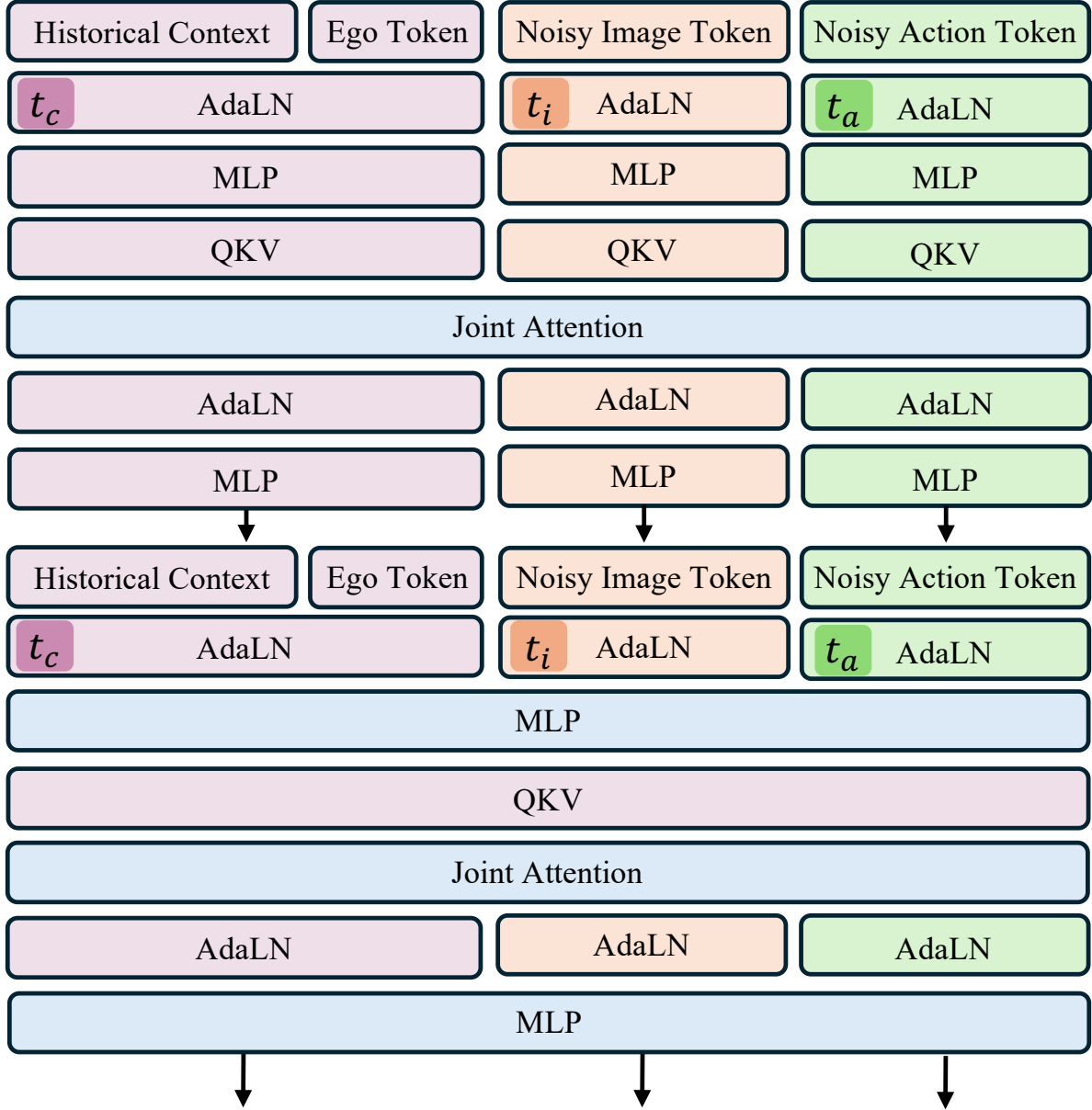


Figure 10: Detailed framework of UniDit. Our architecture is adapted from the existing video generation model (Zhang et al. 2025; Labs et al. 2025), and better inherits its capability of predicting future states.